

Chapter 8: Area and volume

Drill practice and mixed revision of all 13 lessons · 112
questions

NSW syllabus outcome

Stage 4 Measurement and space: MA4-LEN-C-01 (length and
circumference), MA4-ARE-C-01 (area) and MA4-VOL-C-01
(volume and capacity).

CHAPTER

8

Homework. Part 1 is **drill practice**: quick questions on the main skill of each lesson in Chapter 8. Write each answer on its line. Part 2 is **mixed revision**, from **Easy** to **Challenging**. Each question is tagged with its lesson, so you can look back at that lesson's summary if you get stuck.

YOU DO On your own: Part 1, drill practice.

QUESTIONS 1–35

8.01 Units of measurement**Easy** Convert. Big unit to small unit: multiply. Small to big: divide.

1 29 000 mg to g

2 3 t to kg

3 0.13 L to mL

Medium Convert.

4 100 mm to m

5 94 000 mL to L

6 77 mm to cm

7 65 t to kg

8.02 Perimeter**Easy** Find the perimeter of the rectangle.

8 28 m by 13 m

9 3 m by 3 m

10 7 m by 4 m

11 9 m by 7 m

Medium The perimeter and one side of a rectangle are given. Find the other side.

12 P = 64 m, length 28 m

13 P = 24 m, length 11 m

14 P = 70 m, length 25 m

8.03 Circumference of a circle**Easy** Find the circumference, $C = \pi d$. Round to 1 decimal place.

15 $d = 22$ cm

16 $d = 12$ cm

17 $d = 18$ cm

18 $d = 29$ cm

Medium The circumference is given. Find the diameter, $d = C \div \pi$ (1 d.p.).

19 $C = 58$ cm

20 $C = 55$ cm

21 $C = 114$ cm

8.04 Area units**Easy** Convert. $1 \text{ cm}^2 = 100 \text{ mm}^2$, $1 \text{ m}^2 = 10\,000 \text{ cm}^2$, $1 \text{ ha} = 10\,000 \text{ m}^2$.

22 4.5 cm^2 to mm^2

23 0.06 ha to m^2

24 51 km^2 to ha

Medium Convert.

25 $730\,000 \text{ cm}^2$ to m^2

26 $5\,100 \text{ ha}$ to km^2

27 5.1 km^2 to ha

28 $660\,000 \text{ cm}^2$ to m^2

8.05 Area of squares and rectangles**Easy** Find the area of the rectangle, $A = lw$.

29 15 m by 5 m

30 7 m by 2 m

31 12 m by 6 m

32 15 m by 8 m

Medium The area and one side of a rectangle are given. Find the other side.

33 $A = 20 \text{ m}^2$, width 4 m

34 $A = 126 \text{ m}^2$, width 9 m

35 $A = 110 \text{ m}^2$, width 10 m

YOU DO

On your own: Part 1, drill practice.

QUESTIONS 36–62

8.06 Area of a triangle

Easy Find the area of the triangle, $A = \frac{1}{2} \times b \times h$.

36 $b = 8 \text{ cm}, h = 16 \text{ cm}$

37 $b = 18 \text{ cm}, h = 4 \text{ cm}$

38 $b = 16 \text{ cm}, h = 20 \text{ cm}$

Medium The area and the base are given. Find the height, $h = 2A \div b$.

39 $A = 39 \text{ cm}^2, b = 13 \text{ cm}$

40 $A = 117 \text{ cm}^2, b = 18 \text{ cm}$

41 $A = 24 \text{ cm}^2, b = 8 \text{ cm}$

8.07 Area of a parallelogram

Easy Find the area of the parallelogram, $A = bh$.

42 $b = 18 \text{ m}, h = 5 \text{ m}$

43 $b = 5 \text{ m}, h = 11 \text{ m}$

44 $b = 19 \text{ m}, h = 13 \text{ m}$

Medium The area and one measurement are given. Find the other.

45 $A = 120 \text{ m}^2, b = 20 \text{ m}$. Find h .

46 $A = 120 \text{ m}^2, b = 15 \text{ m}$. Find h .

47 $A = 78 \text{ m}^2, b = 6 \text{ m}$. Find h .

8.08 Area of composite shapes

Easy Find the area of the rectangle with a rectangular hole cut out (subtract).

48 $12 \text{ cm} \times 7 \text{ cm}$ rectangle, $4 \text{ cm} \times 4 \text{ cm}$ hole

49 $16 \text{ cm} \times 12 \text{ cm}$ rectangle, $2 \text{ cm} \times 2 \text{ cm}$ hole

8.09 Prisms, cross-sections and views

Easy Name the shape of the cross-section.

50 cube

51 a Toblerone box

52 a can of soup

Medium Is it a prism? Write yes or no.

53 cone

54 triangular prism

55 a brick

8.10 Volume units

Easy Convert. $1 \text{ cm}^3 = 1000 \text{ mm}^3$, $1 \text{ m}^3 = 1\,000\,000 \text{ cm}^3$.

56 $440\,000 \text{ cm}^3$ to m^3

57 $30\,000\,000 \text{ cm}^3$ to m^3

58 $33\,000\,000 \text{ cm}^3$ to m^3

Medium Convert.

59 7.9 cm^3 to mm^3

60 7.5 m^3 to cm^3

61 $58\,000\,000 \text{ cm}^3$ to m^3

62 $29\,000 \text{ mm}^3$ to cm^3

YOU DO

On your own: Part 1, drill practice.

QUESTIONS 63–80

8.11 Volume of a rectangular prism

Easy Find the volume of the rectangular prism, $V = lwh$.

63 $9 \times 4 \times 3$ cm

64 $6 \times 3 \times 9$ cm

65 $4 \times 9 \times 1$ cm

Medium The volume and two measurements are given. Find the third.

66 $V = 140 \text{ cm}^3$, $l = 7$ cm, $w = 10$ cm. Find h .

67 $V = 108 \text{ cm}^3$, $l = 12$ cm, $w = 3$ cm. Find h .

68 $V = 990 \text{ cm}^3$, $l = 11$ cm, $w = 10$ cm. Find h .

8.12 Volume of a prism

Easy Find the volume of the prism, $V = Ah$ (A = area of the cross-section, h = length).

69 $A = 60 \text{ cm}^2$, $h = 8$ cm

70 $A = 42 \text{ cm}^2$, $h = 15$ cm

71 $A = 8 \text{ cm}^2$, $h = 3$ cm

Medium The volume and the area of the cross-section are given. Find the length.

72 $V = 16 \text{ m}^3$, $A = 8 \text{ m}^2$

73 $V = 374 \text{ m}^3$, $A = 34 \text{ m}^2$

74 $V = 77 \text{ m}^3$, $A = 7 \text{ m}^2$

8.13 Capacity

Easy Convert. $1 \text{ L} = 1000 \text{ mL}$, $1 \text{ kL} = 1000 \text{ L}$.

75 27 L to mL

76 0.34 kL to L

77 13 kL to L

Medium Find the capacity of the box in litres.

78 $60 \text{ cm} \times 40 \text{ cm} \times 5 \text{ cm}$

79 $20 \text{ cm} \times 40 \text{ cm} \times 10 \text{ cm}$

80 $40 \text{ cm} \times 20 \text{ cm} \times 15 \text{ cm}$

YOU DO

On your own: Part 2, mixed revision.

QUESTIONS 81–90

81

Easy

Convert.

2.5 L to mL Lesson 8.01

82

Easy

Find the circumference (1 d.p.).

$d = 8$ m Lesson 8.03

83

Easy

Find the area.

rectangle 9 m by 8 m Lesson 8.05

84

Easy

Find the area of the parallelogram.

$b = 13$ m, $h = 5$ m Lesson 8.07

85

Easy

Name the shape of the cross-section.

octagonal prism Lesson 8.09

86

Easy

Find the volume.

7 m $\times$ 4 m $\times$ 3 m Lesson 8.11

87

Easy

Convert.

4 m³ to L Lesson 8.13

88

Easy

Find the perimeter.

rectangle 9 m by 6 m Lesson 8.02

89

Easy

Convert.

6 m² to cm² Lesson 8.04

90

Easy

Find the area of the triangle.

$b = 14$ cm, $h = 6$ cm Lesson 8.06

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 91–100

91 **Easy**

Find the area.

a 12×5 rectangle with a 3×3 hole (cm) Lesson 8.08

92 **Easy**

Convert.

7 cm^3 to mm^3 Lesson 8.10

93 **Easy**

Find the volume.

$A = 22 \text{ m}^2$, $h = 5 \text{ m}$ Lesson 8.12

94

Medium Convert 2750 g to kg. Lesson 8.01

95

Medium Find the circumference of a circle with diameter 9 m (1 d.p.). Lesson 8.03

96

Medium Find the area of a square with side 2.5 m. Lesson 8.05

97

Medium Find the area: base 2.5 m, height 1.6 m. Lesson 8.07

98

Medium Name the cross-section of a hexagonal prism. Lesson 8.09

99

Medium Find the volume of a cube with edge 6 m. Lesson 8.11

100

Medium Find the capacity in litres of a box $25 \text{ cm} \times 20 \text{ cm} \times 10 \text{ cm}$. Lesson 8.13

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 101–110

- 101** **Medium** Find the perimeter of an equilateral triangle with sides 9.5 m. Lesson 8.02

- 103** **Medium** Find the area: base 5 cm, height 3.2 cm. Lesson 8.06

- 105** **Medium** Convert 2 500 000 cm³ to m³. Lesson 8.10

- 107** **Challenging** Convert 3.5 h to minutes. Lesson 8.01

- 109** **Challenging** A rectangle has area 60 m² and width 5 m. Find its length. Lesson 8.05

- 102** **Medium** Convert 30 000 cm² to m². Lesson 8.04

- 104** **Medium** Find the area of an L-shape made from a 6 m × 3 m and a 2 m × 2 m rectangle. Lesson 8.08

- 106** **Medium** Triangle base 5 m, height 4 m, prism length 9 m. Find the volume. Lesson 8.12

- 108** **Challenging** Find the circumference of a circle with radius 3.5 cm (1 d.p.). Lesson 8.03

- 110** **Challenging** A parallelogram has area 50 m² and height 4 m. Find its base. Lesson 8.07

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 111–112

- 111** **Challenging** Sketch the top view of a triangular prism lying on its rectangular face. Lesson 8.09

- 112** **Challenging** A box has volume 72 cm^3 and base area 12 cm^2 . Find its height. Lesson 8.11